EE1-10B MATHEMATICS II

1. *Comments about the actual exam performance follow the solution, in italics. Where there are no comments, the question was done well by most people.*

a) Obtain the Fourier transform of the rectangular pulse function

$$f(t) = \begin{cases} 1, & |t| < T/2, \\ 0, & |t| \geq T/2, \end{cases}$$

where T is a positive constant. [5]

SOLUTION

$$\mathscr{F}[f(t)] = \int_{-\frac{T}{2}}^{\frac{T}{2}} 1 \cdot e^{-i\omega t}\, dt = \left[-\frac{e^{-i\omega t}}{i\omega} \right]_{-\frac{T}{2}}^{\frac{T}{2}} = \frac{1}{i\omega}\left(e^{i\omega\frac{T}{2}} - e^{-i\omega\frac{T}{2}} \right)$$

$$= 2\frac{\sin\left(\omega\frac{T}{2}\right)}{\omega} = T\,\mathrm{sinc}\left(\omega\frac{T}{2}\right),$$

and either form of the answer - sine or sinc - is acceptable.

A few people did not recognize that integration had to be done on the interval $[-T/2, T/2]$ and left the integral with limits $[-\infty, +\infty]$, which was then much harder to solve. It's easier if one looks carefully at the region where $f(t)$ vanishes.

b) Given that a function $g(t)$ has Fourier transform $G(\omega)$, prove the frequency shift property, that is,

$$\mathscr{F}\left[e^{i\omega_0 t} g(t)\right] = G(\omega - \omega_0),$$

where $\mathscr{F}$ denotes the Fourier transform. [5]

SOLUTION

$$\mathscr{F}\left[e^{i\omega_0 t} g(t)\right] = \int_{-\infty}^{+\infty} e^{i\omega_0 t} g(t) e^{-i\omega t}\, dt = \int_{-\infty}^{\infty} g(t) e^{-i(\omega-\omega_0)t}\, dt$$

$$= G(\omega - \omega_0)$$

c) Two planes, Π_1 and Π_2, have cartesian equations

$$-2x+y+2z=1\,,\text{ and }x-2y+z=3\,.$$

Find the intersection of Π_1 and Π_2 and give a geometric interpretation of your answer. [5]

SOLUTION
Solve:

$$\begin{pmatrix}1&-2&1&3\\-2&1&2&1\end{pmatrix}\overset{\mathrm{R}_2+2\mathrm{R}_1}{\sim}\begin{pmatrix}1&-2&1&3\\0&-3&4&7\end{pmatrix}$$

From $\mathrm{R}_2\Rightarrow -3y+4z=7$, so there is one free variable. Let $z=\lambda\in\mathbb{R}$.
Then $y=-\frac{7}{3}+\frac{4}{3}\lambda$ and
$\mathrm{R}_1\rightarrow x=2y-z+3=2(-\frac{7}{3}+\frac{4}{3}\lambda)-\lambda+3=-\frac{5}{3}+\frac{5}{3}\lambda$.

$$\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-\frac{5}{3}\\\frac{7}{3}\\0\end{pmatrix}+\lambda\begin{pmatrix}\frac{5}{3}\\\frac{4}{3}\\1\end{pmatrix}\equiv\begin{pmatrix}-\frac{5}{3}\\\frac{7}{3}\\0\end{pmatrix}+\mu\begin{pmatrix}5\\4\\3\end{pmatrix},$$

the vector equation of a line in $\mathbb{R}^3$.

Some people did not recognize that there was a free variable to be selected. Others did not recognize that they had obtained the equation of a line. Having a system of two equations in three variables has a free variable unless it is impossible. Moreover, it is always geometrically equivalent to the intersection of two planes, which must result in a one-dimensional geometrical object, a line. But mostly done well, a few people had problems with the fractions. A few people gave a cartesian equation of a line in $\mathbb{R}^2$ or even a point.

d) Let

$$A=\begin{pmatrix}\alpha&-1\\1&0\end{pmatrix}.$$

i) Find the values of α such that $A^n=I$, where I is the identity matrix, for $n=3$ and $n=4$. [5]

ii) Prove that A^{-1} exists for all α, but that there is no value of α such that $A=A^{-1}$. [5]

SOLUTION

i)

$$A^2=\begin{pmatrix}\alpha^2-1&-\alpha\\\alpha&-1\end{pmatrix}$$

$$A^3 = \begin{pmatrix} \alpha^3 - 2\alpha & 1-\alpha^2 \\ \alpha^2 - 1 & -\alpha \end{pmatrix} = I_2, \text{ if } \alpha = -1$$

$$A^4 = \begin{pmatrix} \alpha^4 - 3\alpha^2 + 1 & 2\alpha - \alpha^3 \\ \alpha^3 - 2\alpha & 1-\alpha^2 \end{pmatrix} = I_2, \text{ if } \alpha = 0$$

Quite a few errors in A^3 and/or A^4 with people considering only one or two entries in the matrix, rather than finding an α satisfying all entriies It's necessary to do a one-to-one check of all entries of the matrices, i.e. for A^3 to be equal to the identity, one must check that both $(A^3)_{1,1} = (A^3)_{2,2} = 1$ and $(A^3)_{1,2} = (A^3)_{2,1} = 0$ hold true for the same value of α. So in A^4, $\alpha^3 - 2\alpha = 0 \Rightarrow \alpha = 0, \pm\sqrt{2}$ and unless you choose zero, the other options don't work with the other matrix entries. It happens that looking at the $(A^n)_{22}$ element gives unambiguous choices in both cases, but a lot of people didn't see this. Many people got lost in computations. Easy to observe that the second case, i.e. $A^4 = I$, is equivalent to $A^3 = A^{-1}$. Thus, in order to do less work, one might compute A^{-1} from (ii), and use it in (i).

ii)

$$\det A = 1 \neq 0 \Rightarrow A^{-1} \text{ exists } \forall\ \alpha$$

If $A = A^{-1}$ we would have $A^2 = I$, but $(A^2)_{22} = -1$, hence it is impossible $\forall\ \alpha$.

Existence of inverse clear for almost everyone, but the second part proved trickier for many. Most of those that saw it obtained A^{-1} and noted that $(A^{-1})_{11} = 0$, therefore the inverse couldn't be equal to A. Most forgot to mention that whatever they found holds true **for all** α.

2. a) The vector $\underline{\mathbf{y}} \in \mathbb{R}^3$ satisfies

$$\underline{\mathbf{y}} = \underline{\mathbf{a}} + \underline{\mathbf{b}} \times \underline{\mathbf{y}},$$

where $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ are given vectors. By taking the scalar and vector products of this equation with $\underline{\mathbf{b}}$, or otherwise, solve for $\underline{\mathbf{y}}$ in terms of $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$. [6]

SOLUTION

Following the hint,

$$\underline{\mathbf{b}} \cdot \underline{\mathbf{y}} = \underline{\mathbf{b}} \cdot \underline{\mathbf{a}} + \underline{\mathbf{b}} \cdot (\underline{\mathbf{b}} \times \underline{\mathbf{y}}) = \underline{\mathbf{b}} \cdot \underline{\mathbf{a}}$$

as $\underline{\mathbf{b}}$ and $\underline{\mathbf{b}} \times \underline{\mathbf{y}}$ are orthogonal.

$$\underline{\mathbf{b}} \times \underline{\mathbf{y}} = \underline{\mathbf{b}} \times \underline{\mathbf{a}} + \underline{\mathbf{b}} \times (\underline{\mathbf{b}} \times \underline{\mathbf{y}}) = \underline{\mathbf{b}} \times \underline{\mathbf{a}} + (\underline{\mathbf{b}} \cdot \underline{\mathbf{y}})\underline{\mathbf{b}} - (\underline{\mathbf{b}} \cdot \underline{\mathbf{b}})\underline{\mathbf{y}}$$

Now substitute $\underline{\mathbf{b}} \times \underline{\mathbf{y}} = \underline{\mathbf{y}} - \underline{\mathbf{a}}$ and $\underline{\mathbf{b}} \cdot \underline{\mathbf{y}} = \underline{\mathbf{b}} \cdot \underline{\mathbf{a}}$ in the last equation to get

$$\underline{\mathbf{y}} - \underline{\mathbf{a}} = \underline{\mathbf{b}} \times \underline{\mathbf{a}} + (\underline{\mathbf{b}} \cdot \underline{\mathbf{a}})\underline{\mathbf{b}} - (\underline{\mathbf{b}} \cdot \underline{\mathbf{b}})\underline{\mathbf{y}}$$

and solve

$$\underline{\mathbf{y}} = \frac{1}{1 + \underline{\mathbf{b}} \cdot \underline{\mathbf{b}}} (\underline{\mathbf{b}} \times \underline{\mathbf{a}} + (\underline{\mathbf{b}} \cdot \underline{\mathbf{a}})\underline{\mathbf{b}} + \underline{\mathbf{a}})$$

Tough question, very few got this completely. Many got the first part, $\underline{\mathbf{b}} \cdot \underline{\mathbf{y}} = \underline{\mathbf{b}} \cdot \underline{\mathbf{a}}$ *but of these, many then erroneously concluded that this implies* $\underline{\mathbf{y}} = \underline{\mathbf{a}}$*. Many errors in misinterpretation of* $\underline{\mathbf{b}} \times (\underline{\mathbf{b}} \times \underline{\mathbf{y}})$ *as* $(\underline{\mathbf{b}} \times \underline{\mathbf{b}}) \times \underline{\mathbf{y}} = \underline{0} \times \underline{\mathbf{y}}$ *and similar. Careful use of identities is required: triple vector, triple scalar products; although these were on the formula sheet, it's clear that a number of people didn't know how to use them.*

b) Given a matrix and two vectors:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ -1 & 1 & \alpha \end{pmatrix} \quad \text{and} \quad \underline{\mathbf{b}} = \begin{pmatrix} 1 \\ \beta \\ 2 \end{pmatrix} \quad \text{and} \quad \underline{\mathbf{x}} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

i) Find $\det(A)$ and hence, or otherwise, give conditions on α and β so that the linear system of equation with matrix

$$A\underline{\mathbf{x}} = \underline{\mathbf{b}}$$

has (I) no solutions; (II) a unique solution; (III) infinitely many solutions; [5]

ii) For $\alpha = 2$, use Gaussian elimination to obtain the inverse of A. [4]

iii) Solve the system of equations

$$\begin{aligned} -x + y + 2z &= 1 \\ x + y + z &= 2 \\ x + 2y + 3z &= 5. \end{aligned}$$ [2]

SOLUTION

i) Expanding by the first row,
$\det A = 2\alpha - 3 - (\alpha + 3) + (1 + 2) = \alpha - 3$, hence we have a unique solution when $\alpha \neq 3$, case (II). Now if we let $\alpha = 3$, we know there is no unique solution, so either infinitely many, case (III), or no solution at all, case (I). Then do row operations:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & \beta \\ -1 & 1 & 3 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & \beta - 1 \\ 0 & 2 & 4 & 3 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & \beta - 1 \\ 0 & 0 & 0 & 5 - 2\beta \end{array}\right)$$

For $\alpha = 3$, $\begin{cases} \beta = \frac{5}{2} & \text{gives infinitely many solutions} \\ \beta \neq \frac{5}{2} & \text{gives no solutions.} \end{cases}$

(Application of the theorem on existence/uniqueness of solutions. Recall that we have no solution or infinitely many solutions if and only if $\det A \neq 0$. Then, in order for the system to have no solution, the rank of the 3-by-4 matrix has to be not equal to the rank of A, i.e. rank $[A] \neq$ rank $[A|b]$. At the same time, in order for the system to have infinitely many solutions, one must have rank $[A] =$ rank $[A|b]$. It can be easily checked that, whenever $\det A = 0$, one has rk $[A] = 2$. Since the rank is invariant under elementary row operations, one may check rank $[A|b]$ by looking at the $\sim$ computations above. It is then easy to observe that $\beta = 5/2$ yields rank $[A|b] = 2$: infinitely many solutions; and $\beta \neq 5/2$ yields rank $[A|b] = 3$: no solutions.)

A substantial minority did not recognize that, for the system to have a unique solution, $\det A \neq 0$ *is a necessary and sufficient condition. In regards to the existence of no solution or infinitely many solutions, many people made confusing statements.*

Most people got the determinant right, and many used this correctly to obtain the condition for (II), a unique solution. Many errors on condition for no solution/infinitely many solutions. All sorts of permutations of the various necessary statements, e.g. $\beta = 5/2 \Rightarrow$ *no solution, or forgetting that* α *has something to do with this, and so on.*

ii) Row operations on the augmented matrix $(A:I) \sim (I:A^{-1})$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ -1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \begin{array}{l} \mathrm{R}_2 - \mathrm{R}_1 \\ \mathrm{R}_3 + \mathrm{R}_1 \\ \sim \end{array} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 2 & 3 & 1 & 0 & 1 \end{array}\right) \begin{array}{l} R_1 - \mathrm{R}_2 \\ \mathrm{R}_3 - \mathrm{R}_2 \\ \sim \end{array}$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & -1 & 3 & -2 & 1 \end{array}\right) \begin{array}{l} \text{-1}\mathrm{R}_3 \\ \sim \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 2 & -1 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & -3 & 2 & -1 \end{array}\right)$$

$$\begin{array}{l} R_1 + \mathrm{R}_3 \\ \mathrm{R}_2 - 2\mathrm{R}_3 \\ \sim \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & -1 \\ 0 & 1 & 0 & 5 & -3 & 2 \\ 0 & 0 & 1 & -3 & 2 & -1 \end{array}\right)$$

$$\Rightarrow A^{-1} = \begin{pmatrix} -1 & 1 & -1 \\ 5 & -3 & 2 \\ -3 & 2 & -1 \end{pmatrix}$$

Lots of good answers. Problems from arithmetic error, usually. A common mistake was operating on one side, but not on the other. Some people used the cofactor method.

iii) Rewrite as:
$$\begin{aligned} x + y + z &= 2 \\ x + 2y + 3z &= 5 \\ -x + y + 2z &= 1 \end{aligned}$$

$$\Longrightarrow \qquad A\underline{\mathbf{x}} = \underline{\mathbf{b}},\ \underline{\mathbf{x}} = A^{-1} \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} -2+5-1 \\ 10-15+2 \\ -6+10-1 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \\ 3 \end{pmatrix}$$

A large minority did not use the inverse matrix and solved this either by Gaussian elimination or, even more long-winded, adding pairs of equations and eliminating variables. Some used the inverse, but did not notice that it was necessary to rearrange the constants on the r.h.s.

c) Given the symmetric matrix

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

i) Obtain the eigenvalues and normalized eigenvectors of B. [4]

ii) Find an orthogonal diagonalizing matrix P and a diagonal matrix D satisfying

$$B = PDP^T,$$

and without further calculation, write down P^{-1}. [4]

SOLUTION

i)

$$|B-\lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = (1-\lambda)(1-\lambda)-4 = \lambda^2-2\lambda-3 = 0$$

$$\Longrightarrow \lambda = 3,\ -1 \text{ are eigenvalues.}$$

For $\lambda = -1$:

$$(B+I)\underline{\mathbf{x}} = \underline{\mathbf{0}} \quad \Longrightarrow \quad \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$x = -y$, eigenvector: $k\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ $(k \in \mathbb{R})$.

For $\lambda = 3$

$$(B-3I)\underline{\mathbf{x}} = \underline{\mathbf{0}} \quad \Longrightarrow \quad \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$x = y$, eigenvector: $k\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $(k \in \mathbb{R})$.

Normalize: $\hat{\underline{\mathbf{x}}}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $\hat{\underline{\mathbf{x}}}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Mostly done well. Lots of people forgot to normalize. A few zero eigenvectors.

ii) We have distinct eigenvalues and therefore their corresponding eigenvectors are orthogonal.

$P = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ so that $D = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$.

$P^{-1} = P^T = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, as P is an orthogonal matrix.

Mostly done well, a few inverse matrices with zero columns, following from zero eigenvectors. A few forgot that $P^{-1} = P^T$ and had to work a bit to find the inverse.

3. a) Find the general solution of the differential equation

$$\left(2xte^{x^2}+t^2\sin x+\sec^2 x\right)\frac{dx}{dt}=2t\cos x+\cos t-e^{x^2}. \qquad [6]$$

SOLUTION

Rewrite ODE as:

$$\underbrace{\left(2xte^{x^2}+t^2\sin(x)+\sec^2 x\right)}_{P(x,t)}\frac{dx}{dt}\underbrace{+e^{x^2}-2t\cos x-\cos t}_{Q(x,t)}=0$$

This is exact if $\dfrac{\partial P}{\partial t}=\dfrac{\partial Q}{\partial x}$. Check it is the case.

$$\frac{\partial P}{\partial t}=2xe^{x^2}+2t\sin x$$

$$\frac{\partial Q}{\partial x}=2xe^{x^2}+2t\sin x$$

Hence there exists a function $H(x,t)=0$ with: $P=\dfrac{\partial H}{\partial x}$ and $Q=\dfrac{\partial H}{\partial t}$.

$$H=\int P\,dx=te^{x^2}-t^2\cos x+\tan x+f(t)\,,$$

$$H=\int Q\,dt=te^{x^2}-t^2\cos x-\sin t+g(x)\,,$$

where $f(t)$ and $g(x)$ are arbitrary functions. Equate the two to find $f(t)=-\sin t$ and $g(x)=\tan x$. Finally the solution is:

$$te^{x^2}-t^2\cos x+\tan x-\sin t=C,$$

where C is a constant.

A lot of people wrote $\dfrac{dx}{dt}=\dfrac{2t\cos x+\cos t-e^{x^2}}{2xte^{x^2}+t^2\sin x+\sec^2 x}$ *and gave up. Many, many people didn't check for exactness: you must! Oddly, many couldn't integrate* $2xe^{x^2}$ *even among those that had just differentiated* e^{x^2} *a minute earlier. A great many also wrote down*
$H(x,t)=te^{x^2}-t^2\cos x+\tan x-\sin t$, *which is not the solution.*

b) Consider the following second order differential equation:

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 85\cos(3x)\,. \qquad (1)$$

i) Obtain the complementary function for (1). [2]

ii) Obtain a particular integral for (1). [4]

iii) Hence obtain the general solution of the ODE in (1). [1]

SOLUTION

i)

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$$

with auxiliary equation $m^2 - 2m + 2 = 0$, so $m = 1 \pm i$. This gives the complementary function:

$$y_C = e^x\left(A\cos x + B\sin x\right)$$

with A and B arbitrary constants.

Many people wrote $y_C = Ae^{(1+i)x} + Be^{(1-i)x}$ which is formally correct, but not finished. Lots of errors in factorizing the auxiliary equation.

ii) As $\cos(3x)$ is not in the complementary function, it is sufficient to try:

$$\begin{cases} Y_p = C\cos(3x) + D\sin(3x) \\ Y_P' = -3C\sin(3x) + 3D\cos(3x) \\ Y_P'' = -9C\cos(3x) - 9D\sin(3x) \end{cases}$$

Substitute these into the ODE

$$\begin{aligned} -9C\cos(3x) - 9D\sin(3x) - 2\left[-3C\sin(3x) + 3D\cos(3x)\right] \\ +2\left[C\cos(3x) + D\sin(3x)\right] = 85\cos(3x) \end{aligned}$$

and collect the like terms

$$\begin{cases} -7C - 6D = 85 \\ 6C - 7D = 0 \end{cases} \Rightarrow \begin{cases} C = -7 \\ D = -6 \end{cases}$$

The particular integral is then $Y_P = -7\cos(3x) - 6\sin(3x)$.

Mostly arithmetic error, with most of these resulting from substituting $C = 7D/6$ into the first equation and getting messy fractions. Why not use Gaussian elimination? Or multiply the first equation by 6, the second by 7, then add, eliminating C? Few people took this approach. Some took $Y_p = C\cos(3x)$ only.

iii) Hence the general solution is $Y_P + Y_C$:

$$Y = e^x (A\cos x + B\sin x) - 7\cos(3x) - 6\sin(3x)$$

c) Find the solution of the first-order ordinary differential equation

$$t^2 \frac{dx}{dt} = x^2 + xt + t^2,$$

which satisfies the initial condition $x = 1$ when $t = 1$. [6]

SOLUTION

It is a homogeneous equation. Rewrite it as:

$$\frac{dx}{dt} = \left(\frac{x}{t}\right)^2 + \frac{x}{t} + 1.$$

Let $v = \dfrac{x}{t}$, then $x = vt$ and $\dfrac{dx}{dt} = v + t\dfrac{dv}{dt}$ and the ODE becomes:

$$v + t\frac{dv}{dt} = v^2 + v + 1 \quad \Rightarrow \quad t\frac{dv}{dt} = v^2 + 1$$

which is separable.

$$\int \frac{1}{v^2+1} dv = \int \frac{1}{t} dt \quad \Rightarrow \quad \tan^{-1}(v) = \ln|t| + C$$

We can now substitute back: $\tan^{-1}\left(\frac{x}{t}\right) = \ln|t| + c$ and use the initial condition $x(1) = 1$ to find C.

$$\tan^{-1}(1) = \ln(1) + C \quad \Rightarrow \quad C = \frac{\pi}{4}$$

This gives us the solution:

$$\tan^{-1}\left(\frac{x}{t}\right) = \ln|t| + \frac{\pi}{4}$$

and hence

$$x = t\tan\left(\ln|t| + \frac{\pi}{4}\right).$$

A lot of people didn't see the substitution. Others wrote $v = x/t$ and differentiated that, which is more work, and got into a mess. Yet others tried differentiating $v = t/x$, an entirely different substitution, harded to do, and hence getting lost; one person did pull this off, though with a lot of effort! A lot of people did not write x as a function of t, which should always be done when possible.

d) Consider the following first order differential equation:

$$\cos y \frac{dy}{dx} = 3x + \sin y. \qquad (2)$$

i) Use the transformation $u = 3x + \sin y$ to rewrite the differential equation in terms of x and u. [2]

ii) Hence, or otherwise, obtain the solution u to the differerential equation in part (i). [2]

iii) Hence find the solution y of the differential equation (2) which satisfies the initial condition $y = 0$ when $x = 0$. [2]

SOLUTION

i) $u = 3x + \sin y$, so if we take $\dfrac{du}{dx}$ we have:

$$\frac{du}{dx} = 3 + \cos y \frac{dy}{dx} = 3 + u$$

A lot errors began with the odd idea that $\dfrac{du}{dx} = \dfrac{du}{dy}\dfrac{dy}{dx} = \cos y \dfrac{dy}{dx} = u$, *which doesn't work because u is a function of x and y. Others tried to calculate partial derivatives u_x, u_y leading nowhere.*

ii) Integrate $u' = 3 + u$.

$$\int \frac{1}{3+u}\, du = \int 1 \cdot dx \quad \Rightarrow \quad \ln|3+u| = x + C$$

By taking the exponential we have:

$$3 + u = ke^x \quad \Rightarrow \quad u = ke^x - 3$$

A lot of people who got (i) right, proceeded correctly but put the constant of integration too late ending up with $3 + u = e^x + C$ or similar.

iii) Going back to x and y we have

$$3+3x+\sin y = ke^x.$$

The initial condition $y(0)=0$ gives $k=3$ and finally

$$\sin y = 3(e^x - x - 1) \quad \Rightarrow \quad y = \sin^{-1}(3e^x - 3x - 3)$$

Many did not write y as a function of x.

4. a) The function $f(x,t)$ satisfies the wave equation:

$$\frac{\partial^2 f}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}.$$

i) By considering the change of coordinates

$$\rho = x - ct, \qquad \phi = x + ct,$$

and using the chain rule, obtain

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial t}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial t^2}$$

in terms of derivatives with respect to ρ and ϕ. [6]

ii) Use the results of (i) to show that the wave equation in the new variables is

$$\frac{\partial^2 f}{\partial \rho \partial \phi} = 0.$$

[2]

iii) Hence find the general solution of the wave equation for f in terms of x and t. [4]

SOLUTION

i)

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial \rho}\frac{\partial \rho}{\partial x} + \frac{\partial f}{\partial \phi}\frac{\partial \phi}{\partial x} = \frac{\partial f}{\partial \rho} + \frac{\partial f}{\partial \phi} = \left(\frac{\partial}{\partial \rho} + \frac{\partial}{\partial \phi}\right) f$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial \rho}\frac{\partial \rho}{\partial t} + \frac{\partial f}{\partial \phi}\frac{\partial \phi}{\partial t} = -c\frac{\partial f}{\partial \rho} + c\frac{\partial f}{\partial \phi} = c\left(\frac{\partial}{\partial \phi} - \frac{\partial}{\partial \rho}\right) f$$

Hence:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial \rho} + \frac{\partial f}{\partial \phi}\right) = \left(\frac{\partial}{\partial \rho} + \frac{\partial}{\partial \phi}\right)\left(\frac{\partial f}{\partial \rho} + \frac{\partial f}{\partial \phi}\right)$$
$$= \frac{\partial^2 f}{\partial \rho^2} + 2\frac{\partial^2 f}{\partial \rho \partial \phi} + \frac{\partial^2 f}{\partial \phi^2}$$

And similarly:

$$\frac{\partial^2 f}{\partial t^2} = \frac{\partial}{\partial t}\left[c\left(\frac{\partial f}{\partial \phi} - \frac{\partial f}{\partial \rho}\right)\right] = c\left(\frac{\partial}{\partial \phi} - \frac{\partial}{\partial \rho}\right)\left[c\left(\frac{\partial f}{\partial \phi} - \frac{\partial f}{\partial \rho}\right)\right]$$
$$= c^2\left(\frac{\partial^2 f}{\partial \rho^2} - 2\frac{\partial^2 f}{\partial \rho \partial \phi} + \frac{\partial^2 f}{\partial \phi^2}\right)$$

A difficult one for many people, trying to find $\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial \rho} + \frac{\partial f}{\partial \phi}\right)$ *and coming up with many oddities such as* $\frac{\partial^2 f}{\partial x \partial \rho}$ *or* $\left(\frac{\partial f}{\partial \rho}\right)^2$ *or simply* $\frac{\partial^2 f}{\partial \rho^2} + \frac{\partial^2 f}{\partial \phi^2}$*, and similarly for the time-derivative.*

ii) Equate the two expressions

$$\frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} = \frac{c^2}{c^2}\left(\frac{\partial^2 f}{\partial \rho^2} - 2\frac{\partial^2 f}{\partial \rho \partial \phi} + \frac{\partial^2 f}{\partial \phi^2}\right) = \frac{\partial^2 f}{\partial \rho^2} + 2\frac{\partial^2 f}{\partial \rho \partial \phi} + \frac{\partial^2 f}{\partial \phi^2} = \frac{\partial^2 f}{\partial x^2}$$

and cancelling, $\Rightarrow 4\frac{\partial^2 f}{\partial \rho \partial \phi} = 0 \Rightarrow \frac{\partial^2 f}{\partial \rho \partial \phi} = 0$, as required.

A lof of strange outcomes from errors in part (i). Many people tried other approaches to bring in the mixed partial second order derivative which had not materialized in the previous step. Others remembered approximately what it should look like. Many of the errors in (i) and (ii) resulted from mixing up variables: x,t *are dependent on* ϕ,ρ *and vice-versa, so a quantity like* $\frac{\partial^2 f}{\partial x \partial \rho}$ *is meaningless and needs to written in terms of either* x,t *or* ϕ,ρ*.*

iii) Write $\frac{\partial^2 f}{\partial \rho \partial \phi} = 0 \Rightarrow \frac{\partial}{\partial \rho}\left(\frac{\partial f}{\partial \phi}\right) = 0$. If the partial derivative of a function w.r.t. ρ is zero, then that function does not involve ρ and must be a function of ϕ only: $\Rightarrow \frac{\partial f}{\partial \phi} = h(\phi)$, where h is an arbitrary function.

$f = \int h(\phi)\, d\phi = g_1(\phi) + g_2(\rho)$, where g_2 is another arbitrary function. Hence substitute back for ρ, ϕ and obtain
$f = g_1(x+ct) + g_2(x-ct)$, the general solution of the wave equation.

Very few well-justified answers, many just wrote down the result as if it was obvious. Many also used f as one of the arbitrary functions.

b) Let $g(x,y) = (x-y)(xy-1)$.

i) Evaluate the gradient of g and derive its stationary points. [4]

ii) Evaluate the Hessian of g and use it to classify the stationary points of g. Justify your classification. [4]

iii) Evaluate the solutions of $g(x,y) = 0$ and hence, or otherwise, sketch the contours of this function. [5]

SOLUTION

i) $g(x,y) = x^2y - xy^2 + y - x$ and its partial derivatives are:
$g_x = 2xy - y^2 - 1$
$g_y = x^2 - 2xy + 1$.
To obtain the stationary points we set $g_x = 0$ and $g_y = 0$:

$$\begin{cases} (1) & 2xy = y^2 + 1 \\ (2) & 2xy = x^2 + 1 \end{cases} \quad \Rightarrow \quad x^2 = y^2,\ x = \pm y$$

Case $x = y$, substitute into (2) to get:

$$2x^2 = x^2 + 1 \quad \Rightarrow \quad x = \pm 1, y = x = \pm 1$$

The stationary points are at (1,1) and (-1,-1).
Case $x = -y$, substitute into (2) to get:

$$-2x^2 = x^2 + 1 \Rightarrow x^2 = -\frac{1}{3}$$

In this case there are no solutions.
The stationary points are at (1,1) and (-1,-1)

A lot of problems solving equations (1) and (2) , leading to complicated substitutions. Typically, rewriting (1) as $x = \dfrac{y^2+1}{2y}$

and substituting that into (2), which is much harder. Of those that found the simpler $y^2 = x^2$, many stopped with $y = x$ and did not consider $y = -x$. Or did consider it and obtained $(1,-1)$ and $(-1,1)$, or even worse, $x = \pm i/\sqrt{3}$ and thus $y = \mp i/\sqrt{3}$. Oddly, even though they gave these as stationary points they usually didn't classify them in (ii), although many plotted them in the contour sketch, simply ignoring the incompatibility between R^2 and the complex plane. Many people seemed to just guess the right saddle points showing no work. This is of course possible from the contour plot, but only to confirm your results.

ii) Calculate the Hessian matrix:

$$H(g(x,y)) = \begin{pmatrix} g_{xx} & g_{yx} \\ g_{xy} & g_{yy} \end{pmatrix} = \begin{pmatrix} 2y & 2x-2y \\ 2x-2y & -2x \end{pmatrix}$$

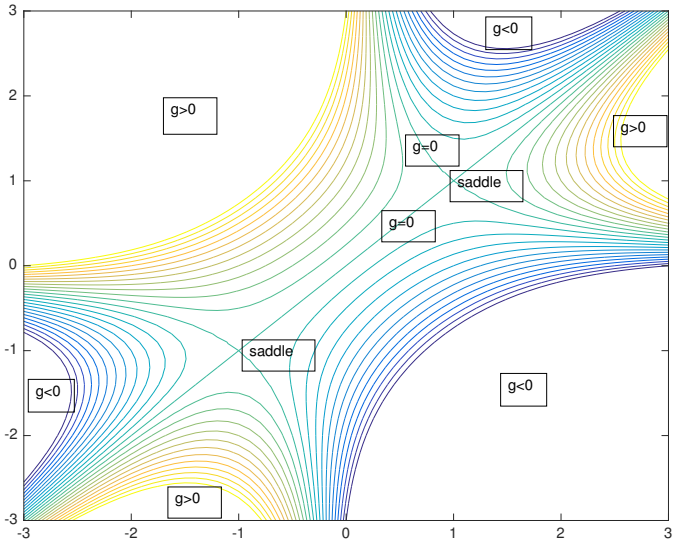
$$H(g(1,1)) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

Since $\det H(g(-1,-1)) = -4 < 0$, (1,1) is a saddle point.

$$H(g(-1,-1)) = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}$$

Since $\det H(g(-1,-1)) = -4 < 0$, (-1,-1) is a saddle point.

iii) $g(x,y) = (x-y)(xy-1) = 0$ implies either $x = y$ or $xy = 1 \Rightarrow y = \dfrac{1}{x}$. These are the contours for $g = 0$ and intersect at the saddle points. Any point where $g \neq 0$ will do to determine the sign elsewhere: $g(1,0) = -1$ so that region has contour lines for $g < 0$ and the contour $g = 0$ gives $g > 0$ on the other side. The figure shows a sketch of the contour plot.



Many good answers, with sloppiness often a problem, or leaving out $g < 0, g > 0$, or the saddle points. Many people just drew the correct contour for $g = 0$ and stopped there. Quite a few drew $y = 1/x$ in the first quadrant only.